

The BAW Ising Machine as a Resonance Computer

Vadde et al. (2026) in the light of Docs. 173, 328 and 343

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The BAW Ising Machine as a Resonance Computer

Abstract

Vadde, Ovcharov, González, Khymyn, Litvinenko and Åkerman (Commun. Phys. 9:290, 8 September 2026, Open Access) present a time-multiplexed Ising machine with 2048 spins on two quartz bulk acoustic wave delay lines. This document reads the work as a case study of the FFGFT classification of resonance computers (Docs. 173, 328, 343): spins as phase states of a ring oscillator, the Barkhausen condition as winding quantisation, parametric $\mathbb{Z}_2$ selection, a coupling regime with a chaos threshold, and — most clearly in quantitative terms — a resolution floor: 99.9 % approximate solutions are found reliably, exact solutions ($E = 0$) are not as problem size grows. All quotations verbatim **[Q]**. Status of the classifications: **[S]**.

Status Markers

[B]	Algebraically proved.
[K]	Numerically confirmed.
[S]	Structurally plausible; physical identification open.
[Q]	Taken from the source (Vadde et al.), quoted verbatim.
[ASSUMPTION]	Axiomatic postulate.

All verbatim quotations are from Vadde et al. (2026), Open Access under CC-BY 4.0; section references follow the structure of that paper.

1 The Source: What Vadde et al. Build

Architecture

[Q]“We use two serially connected 20.5 MHz, 707 μ s bulk acoustic wave delay lines supporting 2048 spins. Our design provides all-to-all connectivity with 15-bit coupling resolution and finds approximate MAX-CUT solutions in 341 ms.” (Abstract)

The spins are not particles but phases of circulating RF pulses:

[Q]“The spin states +1 and −1 are encoded in the relative phase of the pulses, with bistable phases achieved by phase-binarizing the propagating RF pulses using a phase-sensitive amplifier (PSA).” (Section *Design*)

The ring oscillator and the Barkhausen condition

[Q]“The BAW delay line, together with linear amplifiers and the PSA, forms a multi-physics ring oscillator. The key step of the design is to establish a stable circulation of RF pulses within this loop. This requires satisfying the Barkhausen stability criteria: the loop gain must be larger than one, and the total phase shift around the loop must be an integer multiple of 2π .” (Section *Design*)

Parametric binarisation

[Q]“The PSA further narrows the stability criteria to only those signals with either phase 0 or π relative to the pumping signal that is twice the reference frequency.” (Section *Design*)

Minimum cycles and pulse duration

[Q]“Due to the resonance mechanism underlying phase-sensitive amplification, a minimum of 5–10 cycles per pulse is required, setting a lower bound on the pulse duration to ~ 250 –500 ns. This constraint dominates and effectively determines the pulse period.” (Section *Design*)

Coupling strength and chaos threshold

[Q]“The amplitude of the coupling signal can be tuned using the potentiometer of the DAC, allowing an additional control to ensure the injected signal remains within 5–30 % of the circulating RF signal and does not overpower it, avoiding chaotization of the oscillatory circuit.” (Section *Design*)

[Q]“The performance of BAWIM depends on the coupling strength, and there exists an optimal coupling strength depending on the problem instance.” (Section *Performance*)

Approximation versus exactness

MAX-CUT:

[Q]“Here, we observe that BAWIM attains up to 95 % of HbSB’s best performance in terms of Ising energy, and up to 99.99 % in terms of the MAX-CUT score.” (Section *Dependence on the edge density*)

Number partitioning:

[Q]“Here, the exact solution corresponds to a difference of 0, while the 99.9 % approximate solution indicates that the difference is less than 0.1 % of the maximum possible value [...]. In all cases, BAWIM outperforms the HbSB algorithm, achieving a 100 % success rate in most instances when considering the 99.9 % approximate solution.” (Section *Number partitioning*)

Figure 5(b) shows what the text does not spell out: the curve “BAWIM Exact” falls from about 90 % success at 32 numbers to near zero at 2048 numbers, while “BAWIM 99.9 %” stays at 100 % across the whole range.

Sudoku:

[Q]“Notably, even when the Ising energy approaches the ground state, the Sudoku solution remains far from correct. [...] Sudoku requires exact constraint satisfaction; therefore, a near-ground-state Ising energy may still correspond to an invalid solution.” (Section *Sudoku*)

Thermal stability

[Q]“In contrast, our BAWIM exhibits a temperature coefficient of phase accumulation of only 780 deg/°C, which is 2.21×10^4 times lower than that of the CIM.” (Section *Comparison with CIMs*)

2 Classification 1: Ising Machine = Resonator (Doc. 173)

Doc. 173 §*The Arc Closes* says of the coherent Ising machine: “This resonator also has discrete states: it oscillates in either one or the other phase — spin-up or spin-down, zero or one. The excitation is analogue — continuous laser light, continuously adjustable coupling strengths. The state is discrete.” And in the conclusion: “An Ising machine is a resonator. All discrete from inside. All analogue from outside.”

BAWIM is this description in its purest form — no photonics, no cryogenics. The states are phases 0 and π ; the excitation is a continuous RF carrier; the coupling is an analogue current with 15-bit resolution. The spin is not an object but a phase relation. **[S]**

3 Classification 2: Barkhausen as Winding Quantisation (Doc. 314)

The Barkhausen condition “integer multiple of 2π ” is a winding number: only states whose accumulated phase around the closed loop is $2\pi n$, $n \in \mathbb{Z}$, circulate stably. **[B]** (Doc. 314, Ch. D2): “winding numbers are topological”; the mode structure on T^4 is fixed by integer winding around every cycle.

The ring oscillator is an S^1 — a single cycle. The Barkhausen condition is winding quantisation on that S^1 . BAWIM thus realises technically what FFGFT posits for T^4 : states exist only on integer windings. **[S]**

4 Classification 3: Parametric $\mathbb{Z}_2$ Selection (Doc. 339)

The PSA is pumped at 2ω and passes only phases 0 and π . This is period doubling: the phase state space S^1 is reduced to two points $\{0, \pi\} \cong \{+1, -1\}$.

[B] (Doc. 339, §3.1): The fixed points $\{+1, -1\}$ of the Frobenius on $GF(27)^*$ form the massive $\mathbb{Z}_2$ sector. **[B]** (Doc. 336): The norm classes $\{+1, -1\}$ separate light from heavy.

BAWIM’s parametric selection is a technical realisation of the same dichotomy: from a continuous phase space exactly two fixed points are filtered out. That there are two and not three is due to pumping at 2ω ; pumping at 3ω would yield a $\mathbb{Z}_3$ selection $\{0, 2\pi/3, 4\pi/3\}$ — the Potts machine that Whitehead et al. (Ref. 12 in Vadde et al.) built in CMOS. **[S]**

5 Classification 4: Coupling Regime with Chaos Threshold (Doc. 328)

Doc. 328 treats coupling regimes of coupled oscillators: the Adler model, synchronisation, phase locking, and the limit beyond which coupling destroys the oscillator. BAWIM sets exactly this limit operationally: injection at 5–30 % of the circulating signal, “avoiding chaotization”.

The existence of an *optimal* coupling strength per problem instance (Suppl. Note 5) is the Adler sweet spot: too weak — no synchronisation, the system stays in random states; too strong — chaotisation, phase binarisation breaks down. In between lies the window in which the oscillator bank settles into the energy minimum. **[S]**

6 Classification 5: The Resolution Floor (Doc. 343 §G’)

This is the classification with the clearest quantitative support. Doc. 343 §G’ states: a resonance-based computer is capped at $N_{\max} \approx 1/\varepsilon$, where ε is the coupling tolerance; it finds minima to tolerance but not the exact zero. **[B]**

BAWIM displays this behaviour in three forms:

MAX-CUT. 95 % of Ising energy, 99.99 % of cut value. The optimum is reached to within fractions of a percent — but not exactly.

Number partitioning. The difference $E = |\sum_A a_i - \sum_B a_i|$ is brought below 0.1 % of the total sum with 100 % success. The *exact* partition $E = 0$, by contrast, is still found in ~90 % of runs at 32 numbers and practically never at 2048 numbers (Fig. 5b). Coupling resolution is 15 bit; the integers are ≤ 178 ; the total sum at 2048 numbers is $\sim 1.8 \times 10^5$. A tolerance of

0.1 % corresponds to ~ 180 — the magnitude of a single number. The machine solves the problem to within one element, not to zero.

Sudoku. At 99.42 % of ground-state energy the solution is still invalid. Sudoku is a constraint-satisfaction problem: there is no “almost right”. Vadde et al. say so themselves: “a near-ground-state Ising energy may still correspond to an invalid solution.” That BAWIM nevertheless solves the Sudoku is due to the smallness of the problem (729 spins) and to the ground-state energy being an isolated minimum here; HbSB fails on the same problem.

The classification. All three findings are consistent with Doc. 343: the resonance computer is an analogue computer with a resolution floor. It is *not* worse than a digital algorithm — on approximations it is better (NPP 99.9 %: BAWIM beats HbSB throughout). It is *different*: it finds the neighbourhood of the minimum, not the point. For optimisation that is the right task; for exact arithmetic it is not. **[S]**

7 What Does Not Correspond

Classical, not quantum. BAWIM is a classical RF ring oscillator at room temperature. It claims no quantum advantage and asserts none. The FFGFT classifications concern resonator structure, not quantum coherence.

One cycle, not four. The ring oscillator is S^1 ; T^4 has four independent cycles. The winding parallel (Classification 2) holds for one cycle; the T^4 -specific statements (fixed points, $\mathbb{Z}_3$ triality, mass assignment) have no counterpart in BAWIM.

Benchmark caveat. Vadde et al. stress twice that no parameter optimisation was performed for either HbSB or BAWIM. The quantitative comparisons (341 ms versus 102 ms; NPP success rates) are therefore snapshots. The *qualitative* structure — approximation reliable, exactness scale-dependent — is unaffected.

The resolution floor here is technical, not fundamental. The 15-bit coupling resolution is an electronics choice (AD9708 DAC), not a law of nature. Doc. 343 §G’ speaks of $\varepsilon = \xi$ as the natural floor; BAWIM’s $2^{-15} \approx 3 \times 10^{-5}$ happens to lie in the same order of magnitude as $\xi = 1.33 \times 10^{-4}$, without any connection being claimed here.

8 Summary

BAWIM (Vadde et al.)	FFGFT	Doc.	Status
spin = phase $0/\pi$ of a pulse	Ising machine = resonator	173	[S]
Barkhausen: phase = $2\pi n$	winding quantisation on S^1	314	[S]
PSA at 2ω : $\{0, \pi\}$	$\mathbb{Z}_2$ fixed points $\{+1, -1\}$	339, 336	[S]
coupling 5–30 %, chaos limit	Adler regime, coupling threshold	328	[S]
99.9 % yes, exact no	resolution floor $N_{\max} \approx 1/\varepsilon$	343 §G’	[S]

BAWIM is the clearest published case to date of a resonance computer in the sense of Docs. 173/343: thermally stable, without post-selection, without cryogenics, with reproducible approximate solutions and a measurable resolution floor for exact ones. It confirms the FFGFT classification not as a theory but as a taxonomy: what resonates finds minima — and finds them to tolerance.

Source. Vadde, V.; Ovcharov, R.; González, V. H.; Khymyn, R.; Litvinenko, A.; Åkerman, J.: *A 2048-spin bulk acoustic wave Ising machine for number partitioning and Sudoku*. Commun. Phys. **9**, 290 (2026). DOI: 10.1038/s42005-026-02846-7. Open Access, CC-BY 4.0. Received 2 March 2026, accepted 20 August 2026, published 8 September 2026.